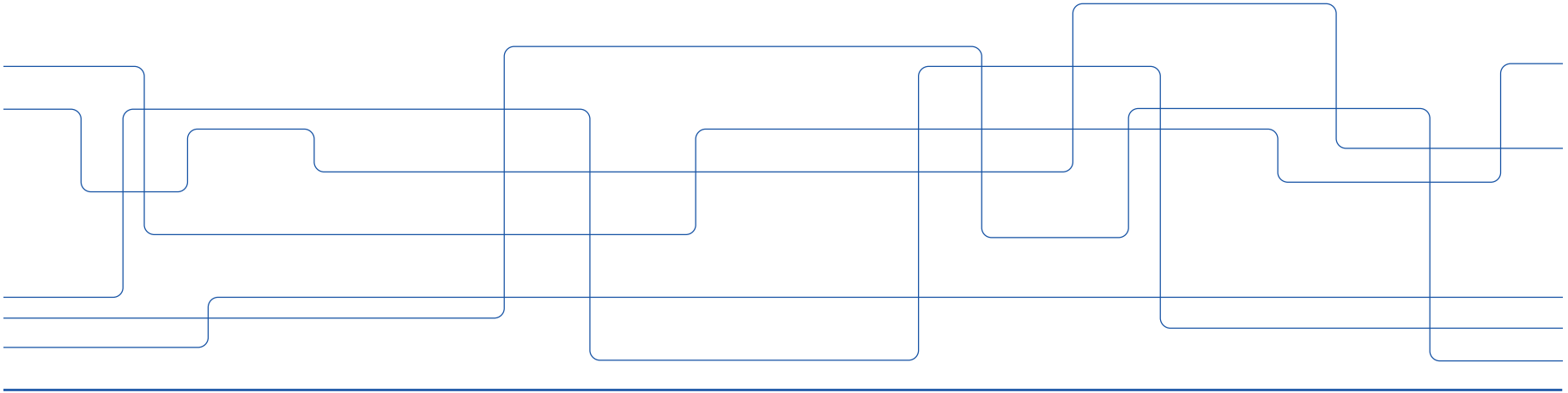
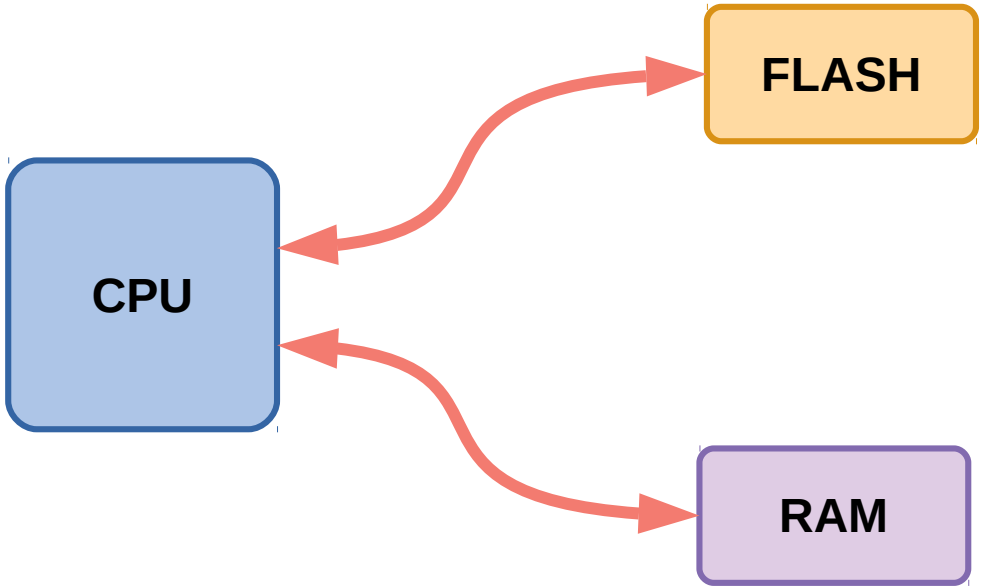
Direct Memory Access

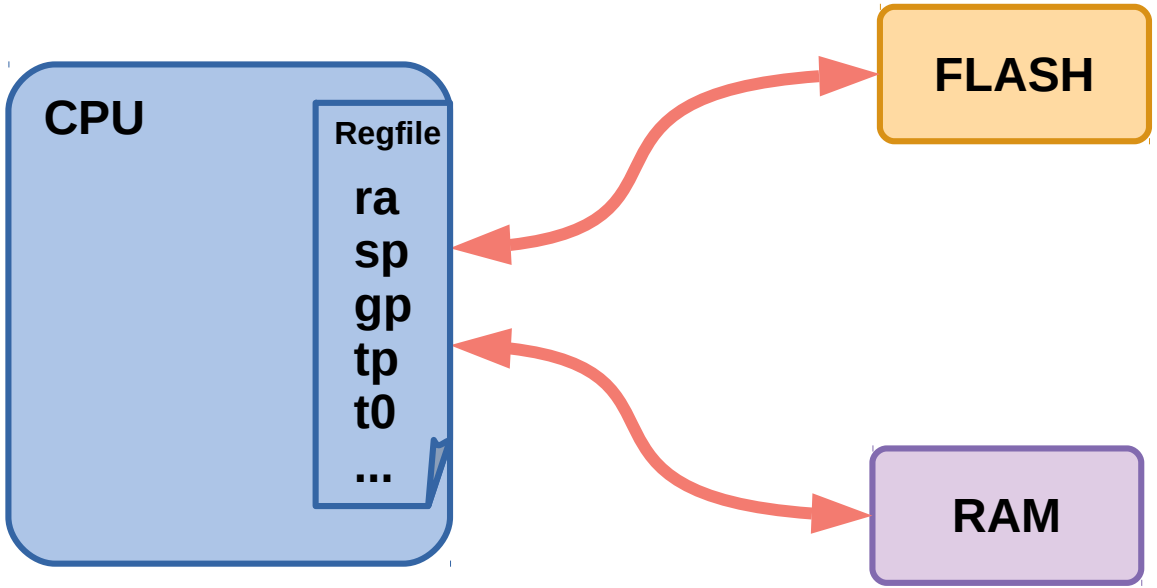
Introduction



Minneshantering i ett datorsystem



Minneshantering i ett datorsystem



Minneshantering i ett datorsystem

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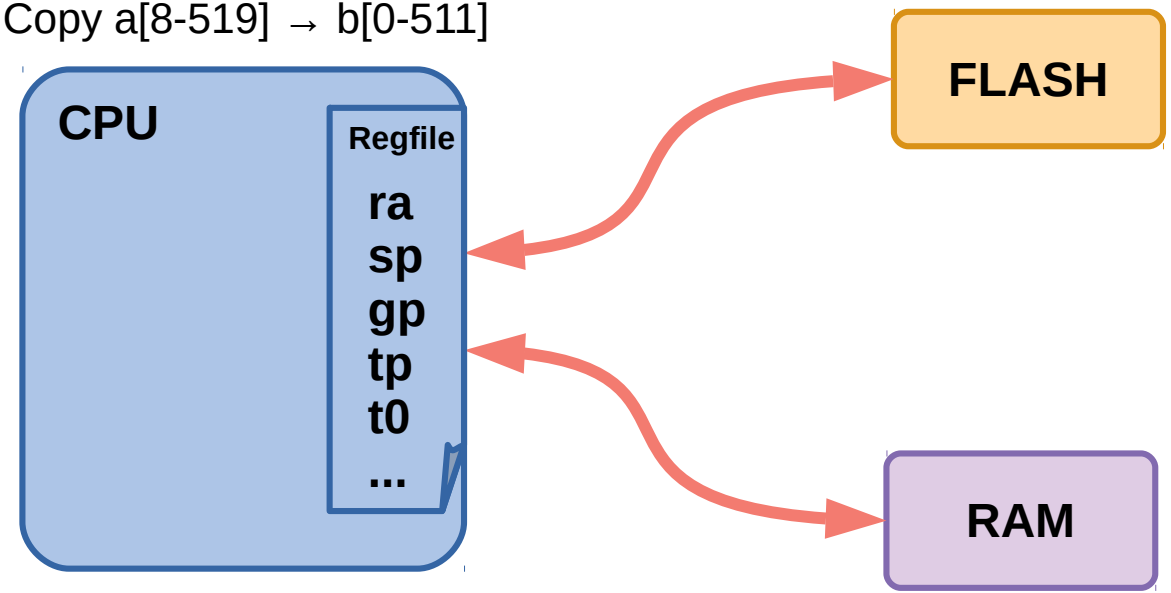

Minneshantering i ett datorsystem

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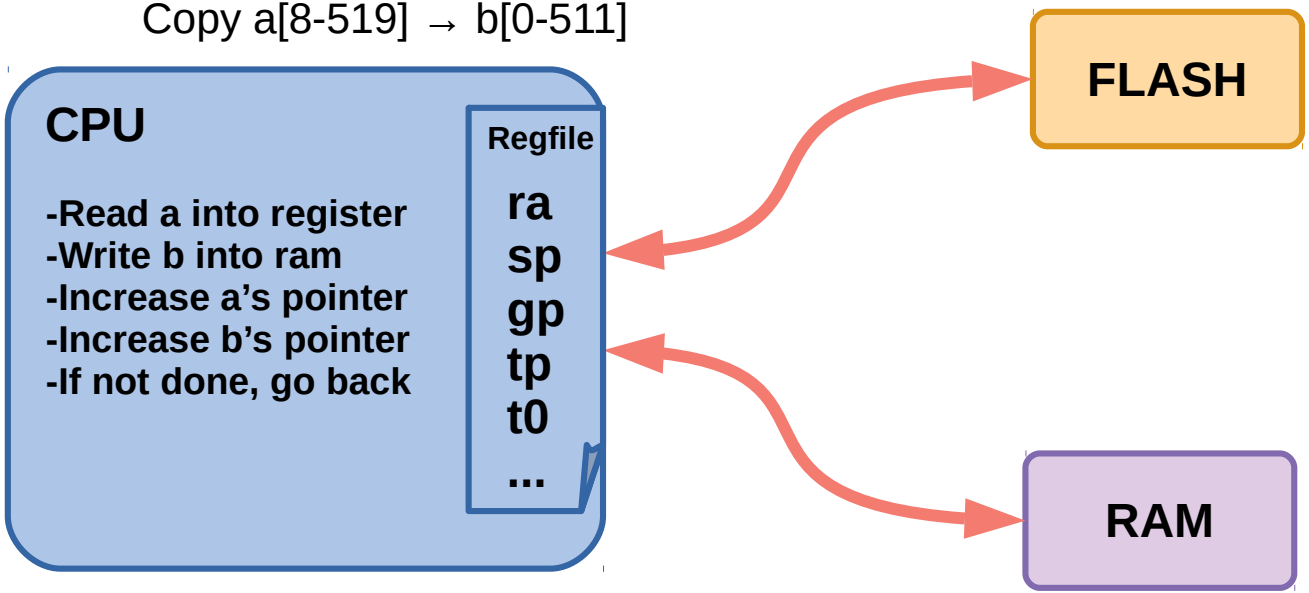

Minneshantering i ett datorsystem

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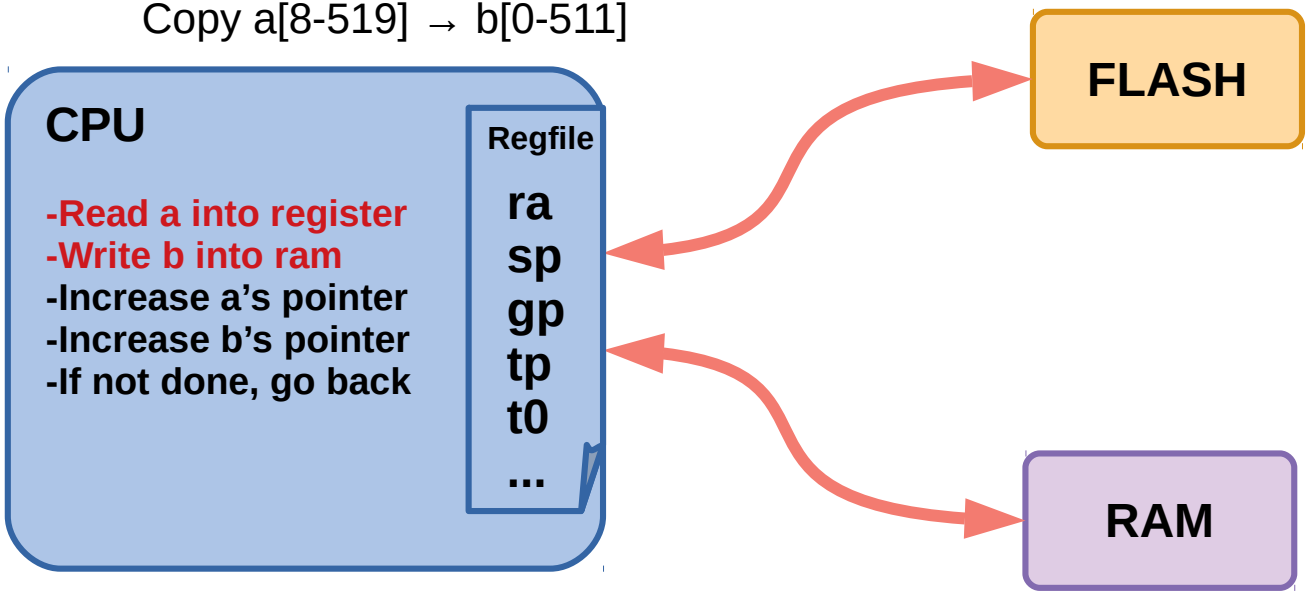

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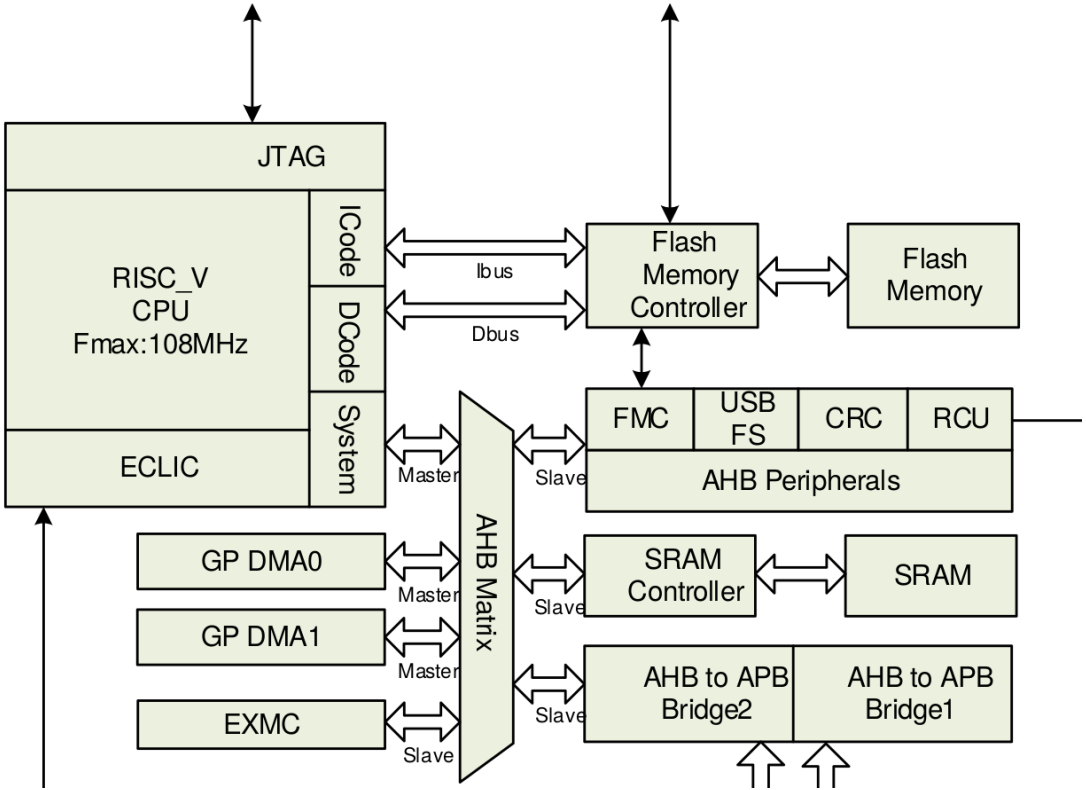
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Fmax:108MHz
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DCode
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ECLIC
Ibus
Dbus
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Memory
Controller
Flash
Memory
FMC
USB
FS
CRC
RCU
AHB Peripherals
Master
Slave
AHB Matrix
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GP DMA1
EXMC
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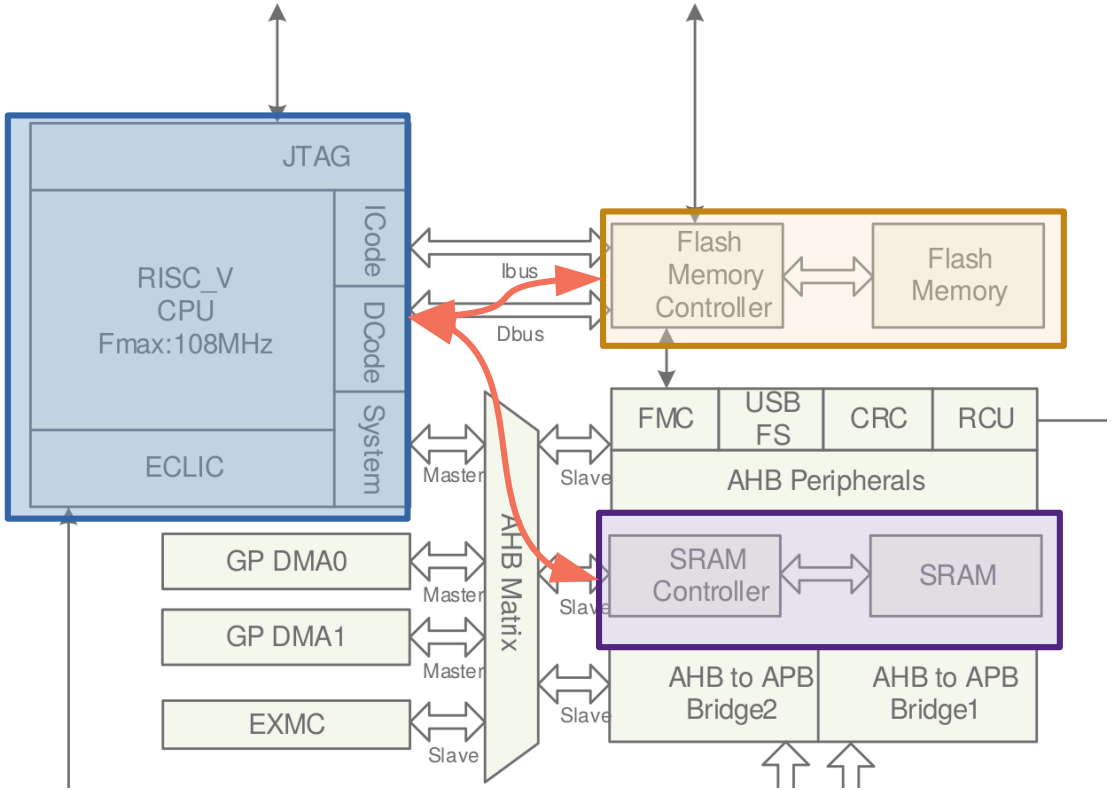
Controller
SRAM
AHB to APB
Bridge2
AHB to APB
Bridge1

I GD32VF103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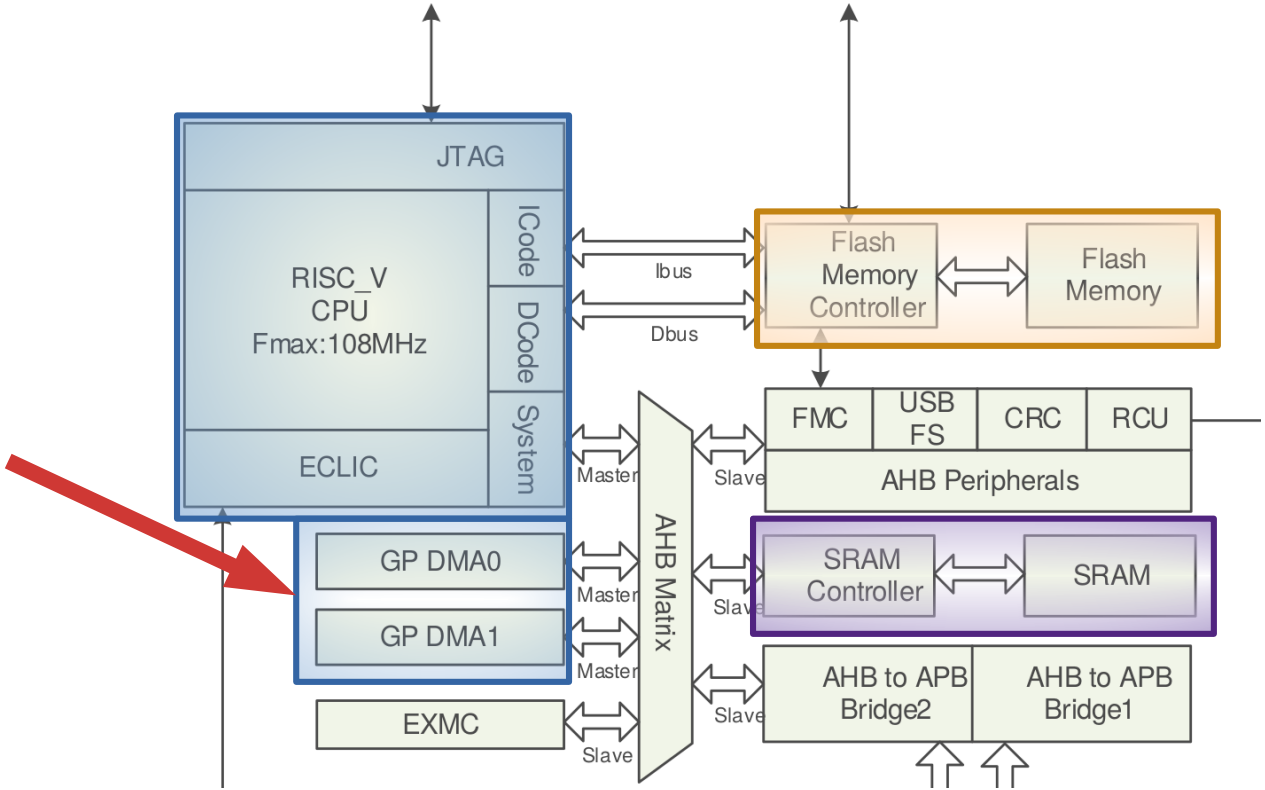
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9. Direct memory access controller (DMA)

9.1. Overview

The direct memory access (DMA) controller provides a hardware method of transferring data between peripherals and/or memory without intervention from the CPU...

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The direct memory access (DMA) controller provides a hardware method of transferring data between peripherals and/or memory **without intervention from the CPU...**

DMA

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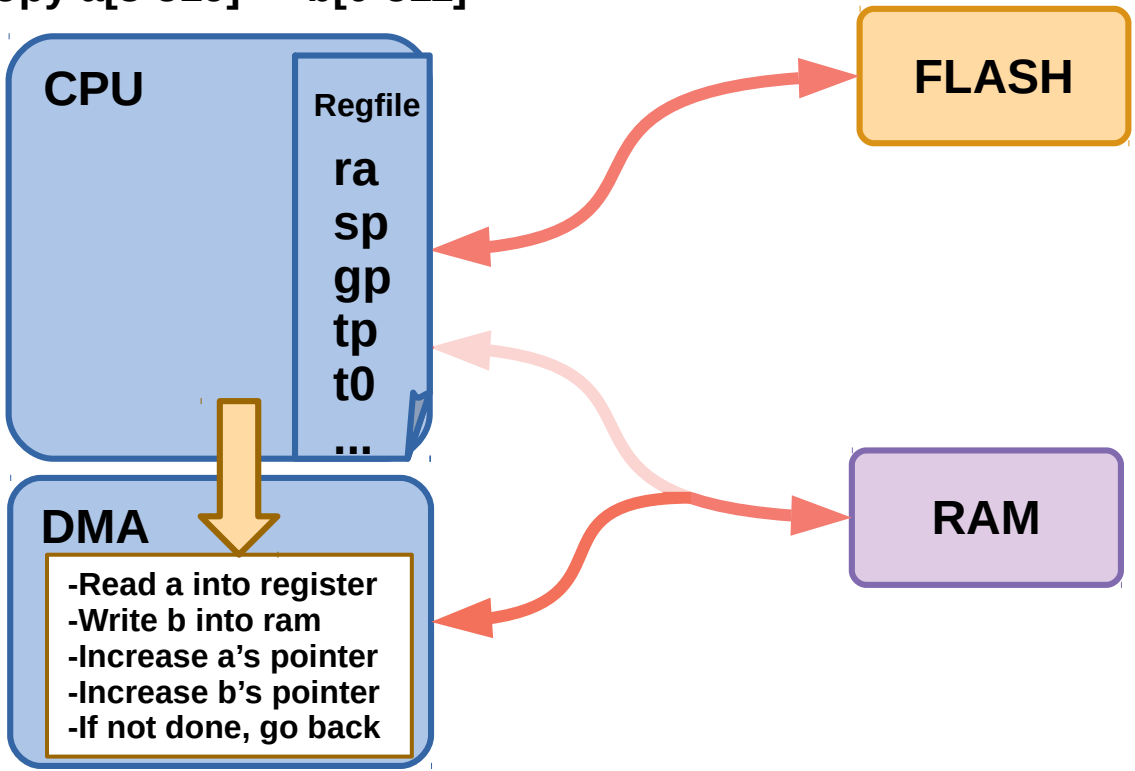

DMA

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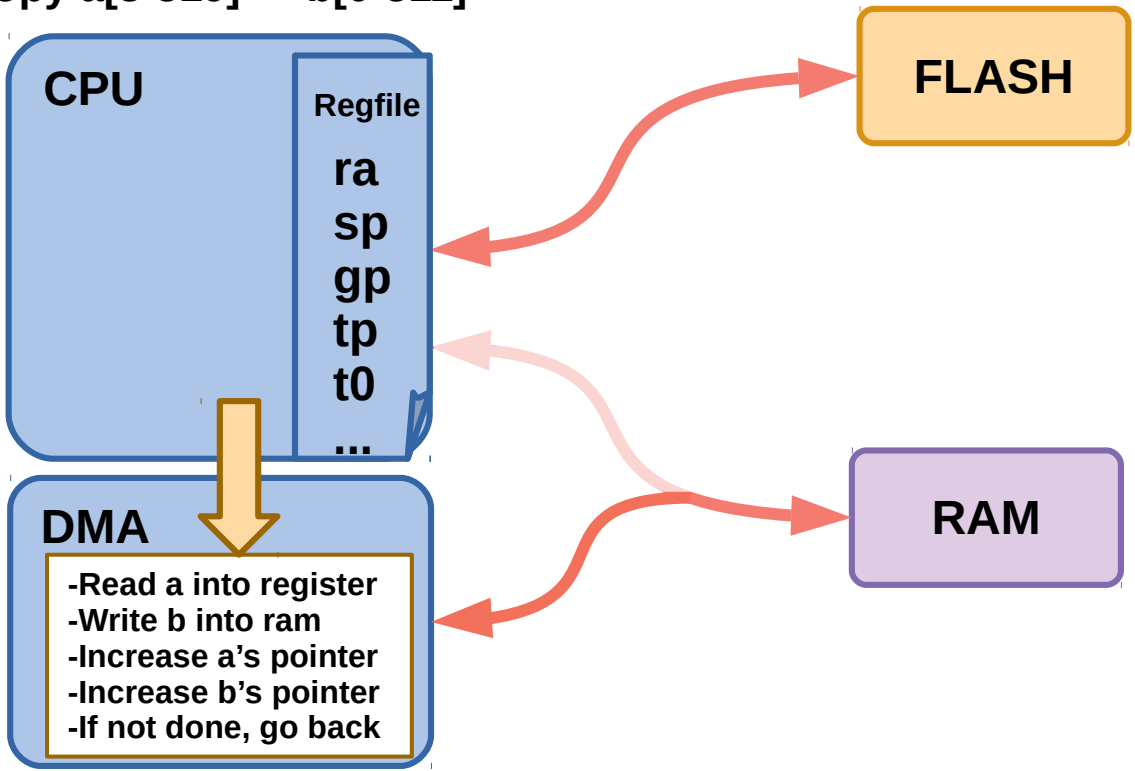

DMA

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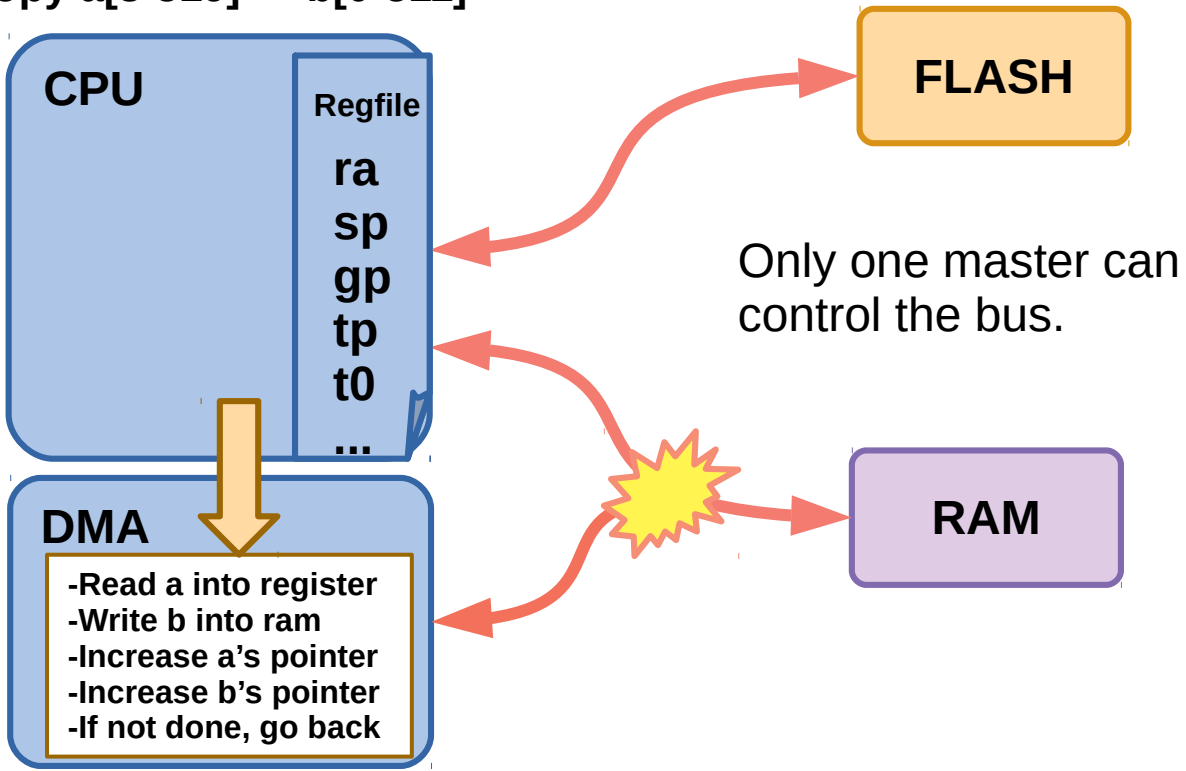

Bus control

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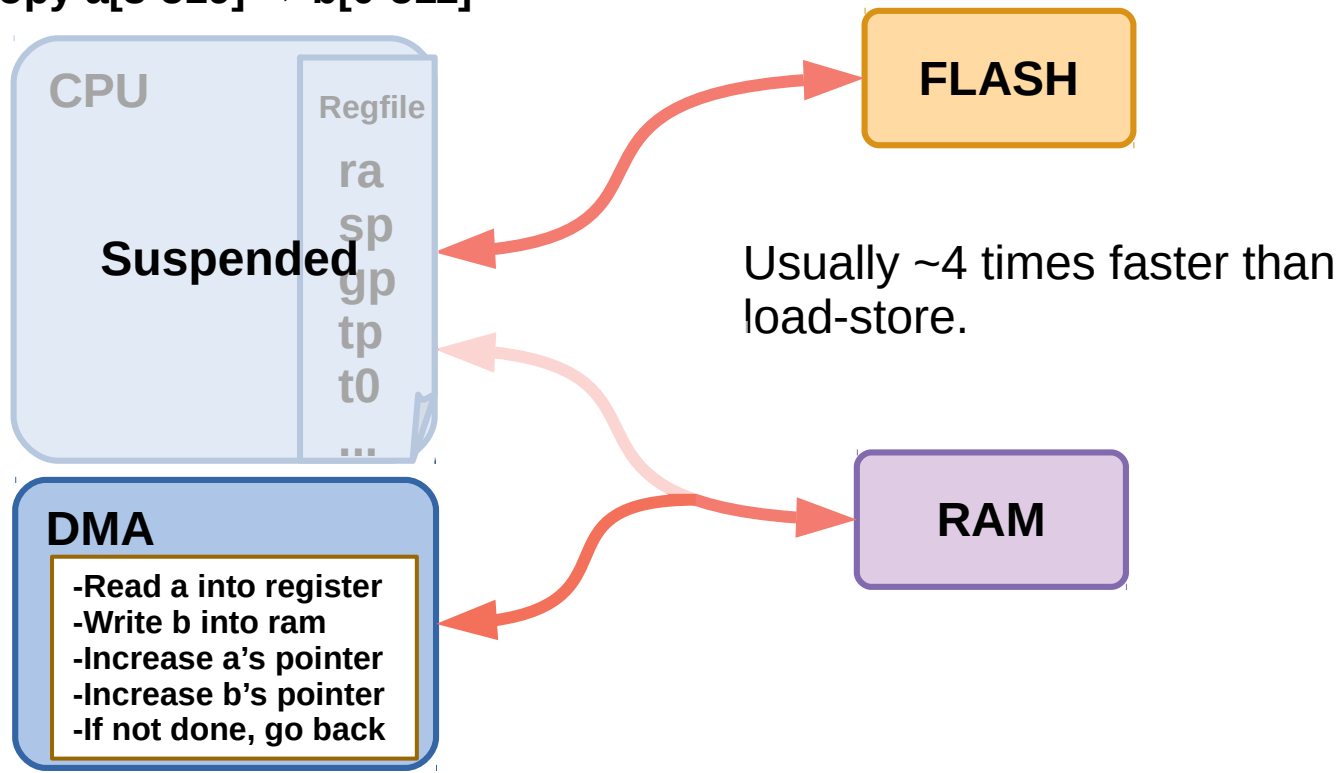
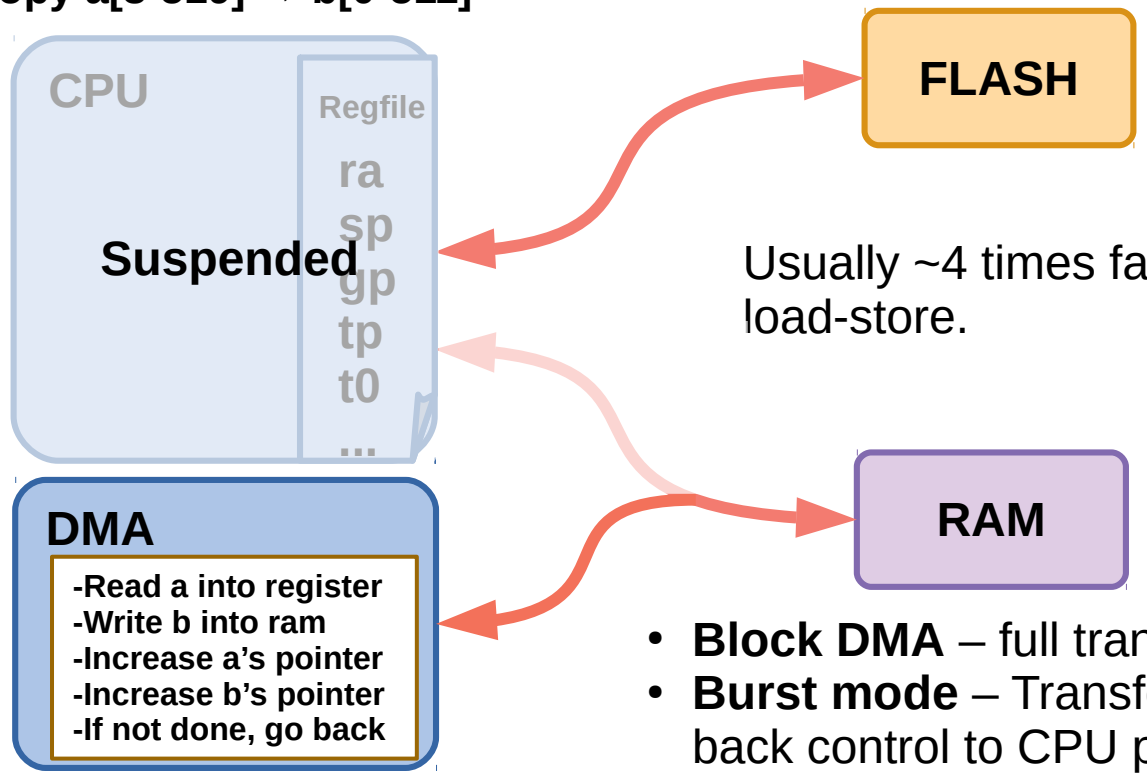


Bus control

Copy a[8-519] → b[0-511]



Usually ~4 times faster than load-store.

- **Block DMA** – full transfer in one go
- **Burst mode** – Transfer a bit then give back control to CPU periodically

Bus control – cycle stealing

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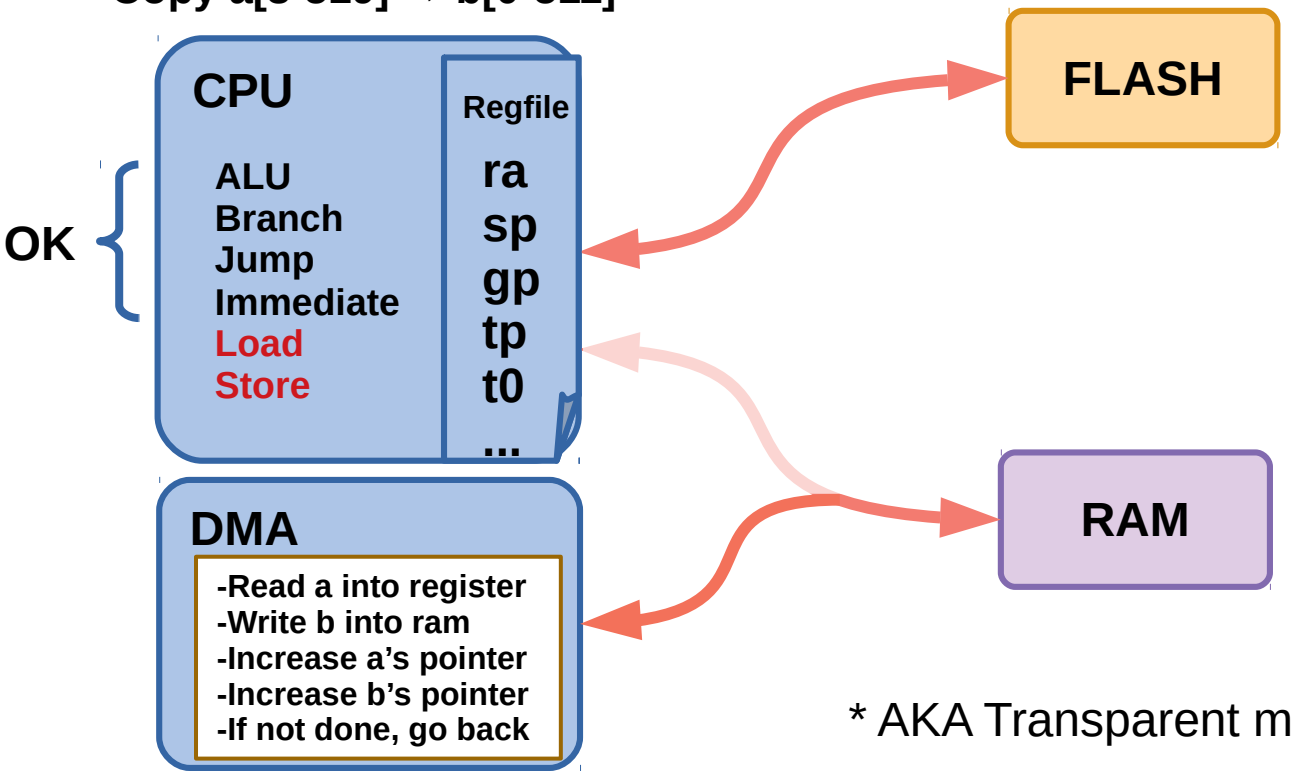

* AKA Transparent mode

Bus control – cycle stealing

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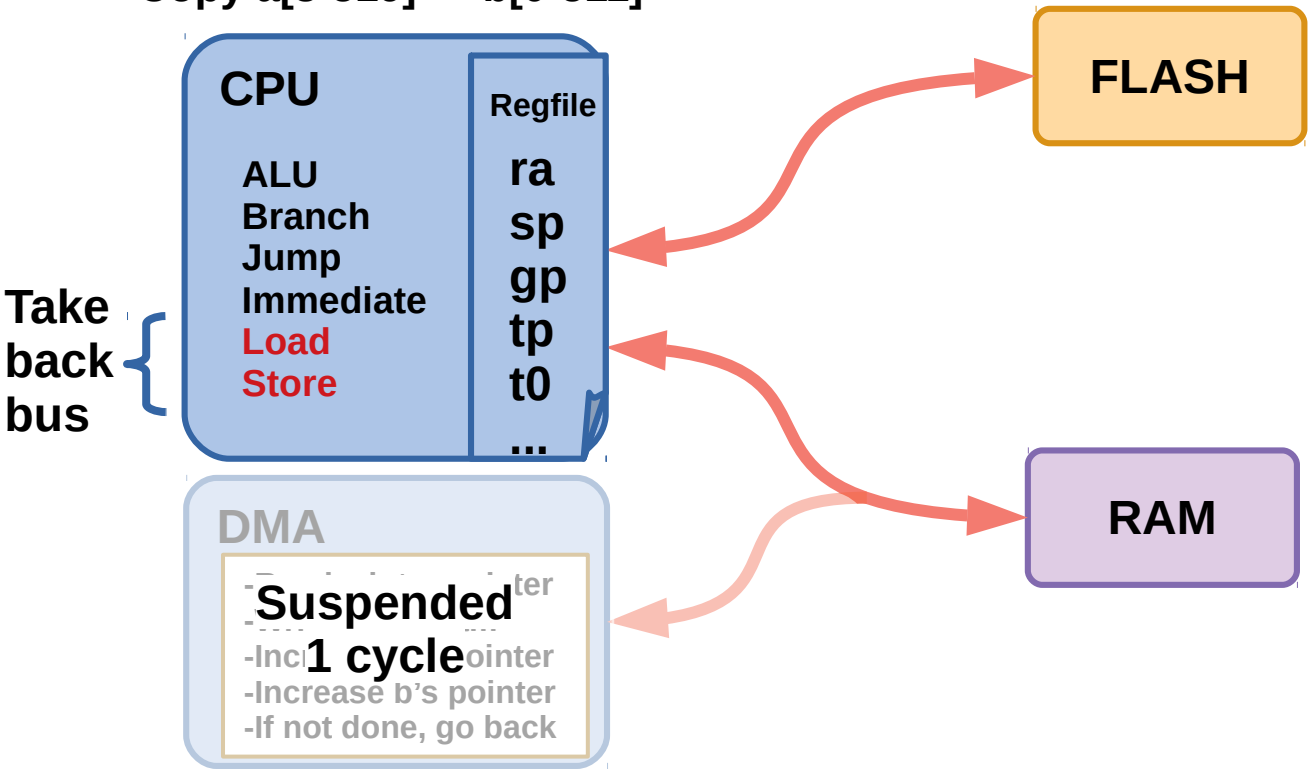

Bus control – cycle stealing

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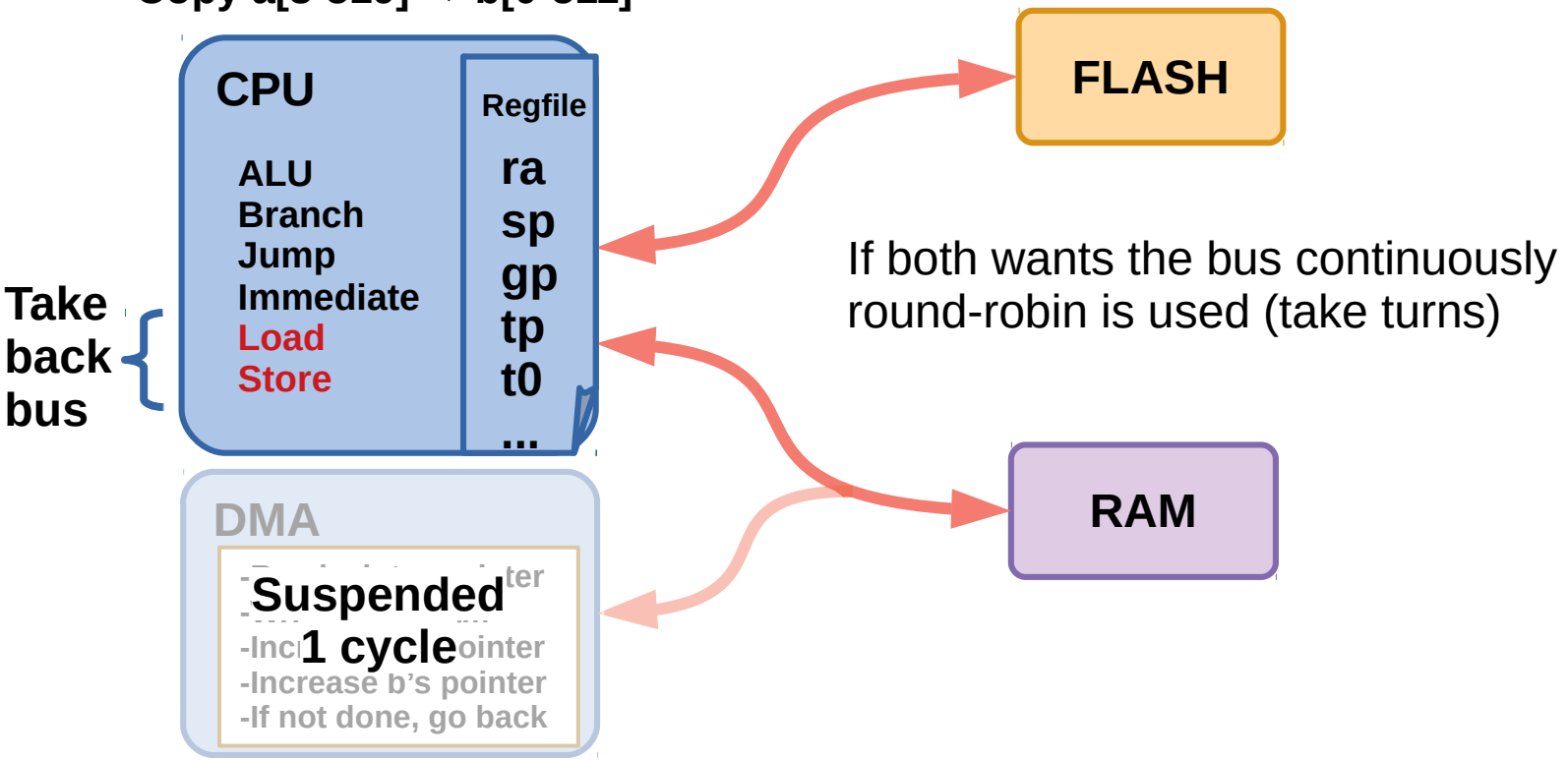
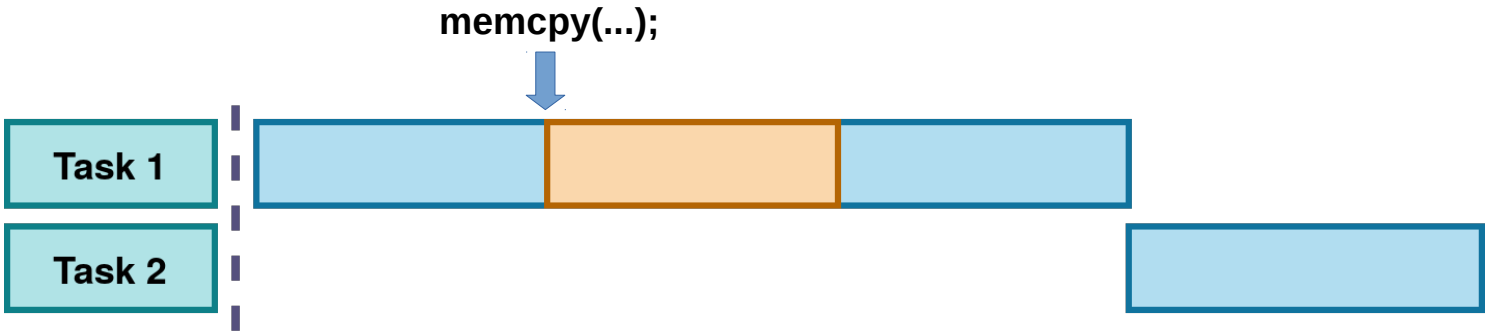
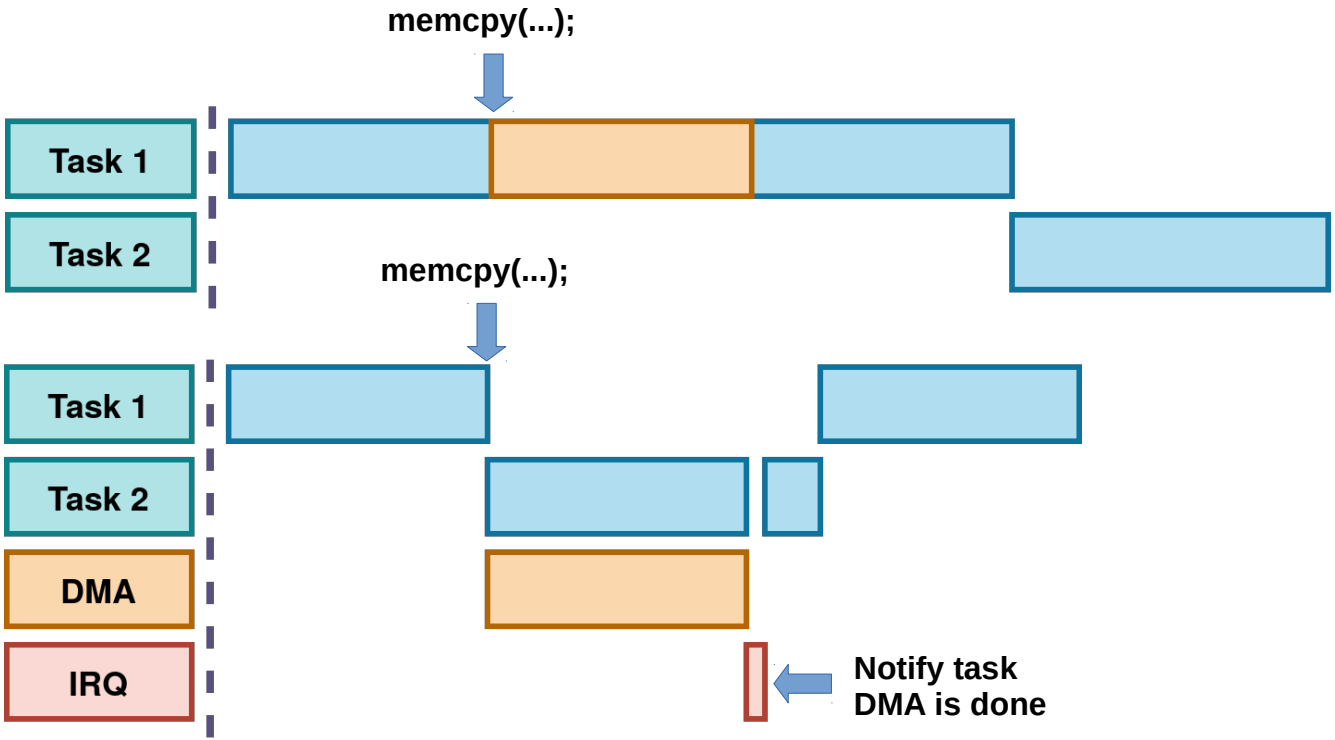


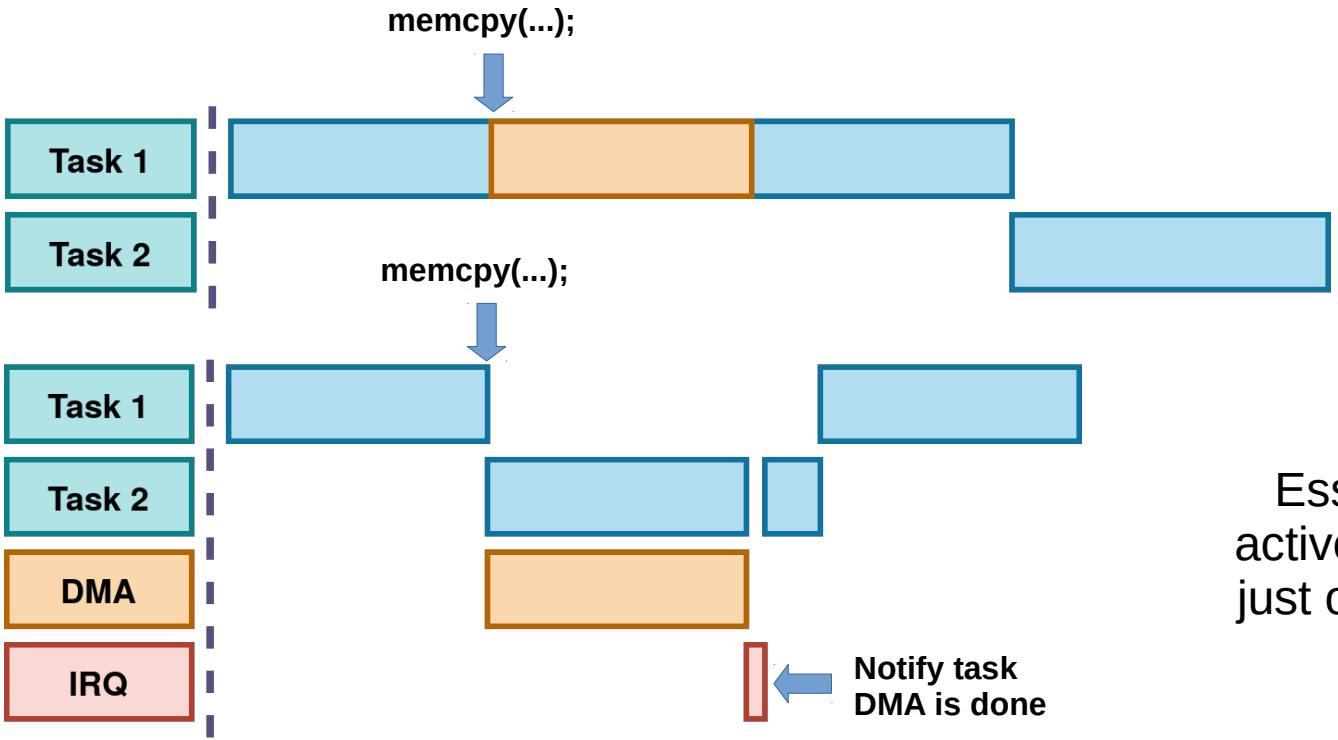
In a multitasking environment



In a multitasking environment



In a multitasking environment



Essentially two active threads with just one processor core!

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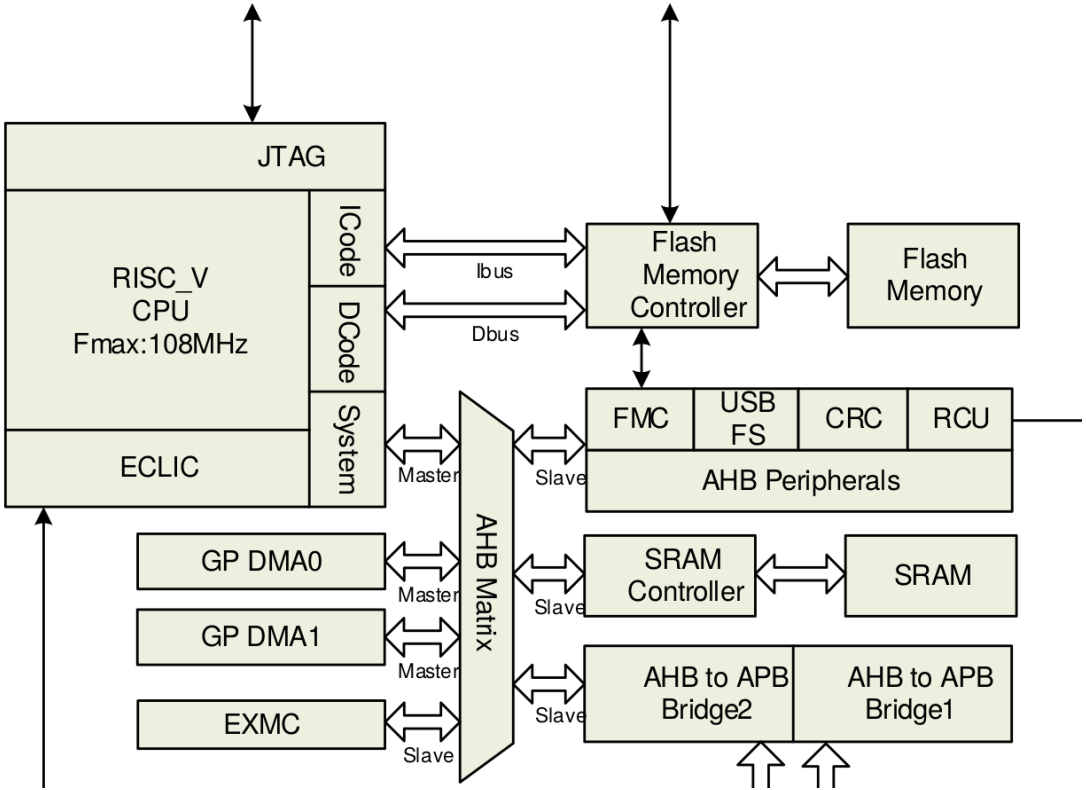
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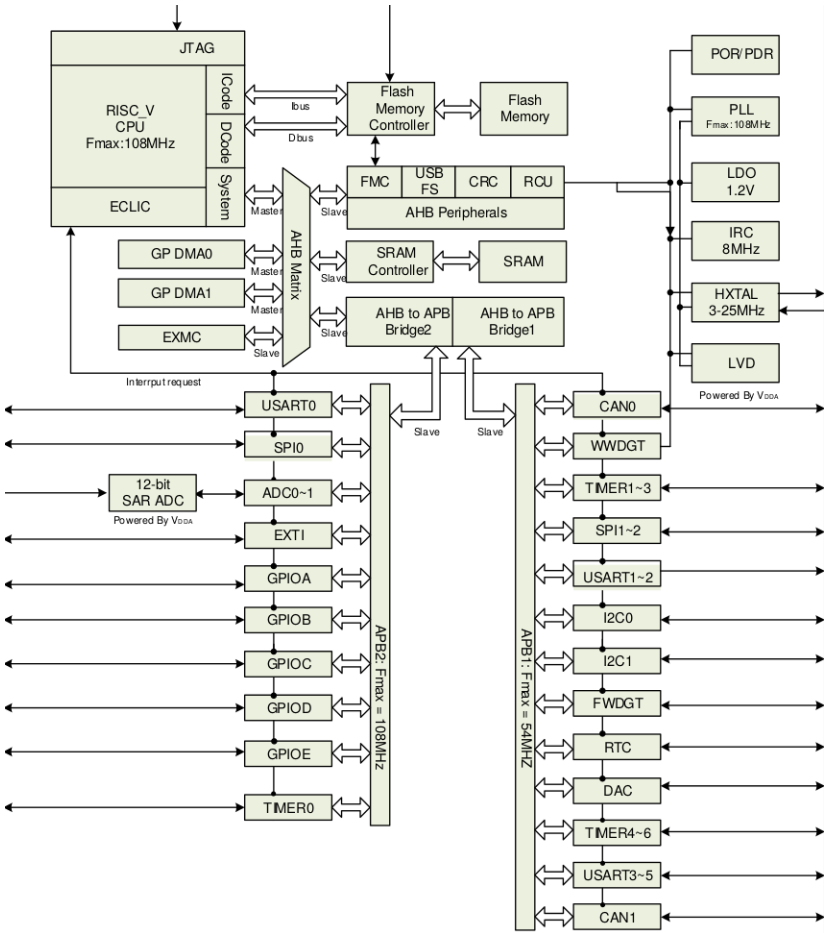
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GP DMA1
EXMC
SRAM Controller
SRAM
AHB to APB Bridge2
AHB to APB Bridge1
Interrput request
POR/PDR
PLL
Fmax:108MHz
LDO 1.2V
IRC 8MHz
HXTAL 3-25MHz
LVD
Powered By VDDA
USART0
SPI0
12-bit SAR ADC
ADC0~1
EXTI
GPIOA
GPIOB
GPIOC
GPIOD
GPIOE
TIMER0
APB2: Fmax = 108MHz
CAN0
WWDGT
TIMER1~3
SPI1~2
USART1~2
I2C0
I2C1
FWDGT
RTC
DAC
TIMER4~6
USART3~5
CAN1
APB1: Fmax = 54MHz

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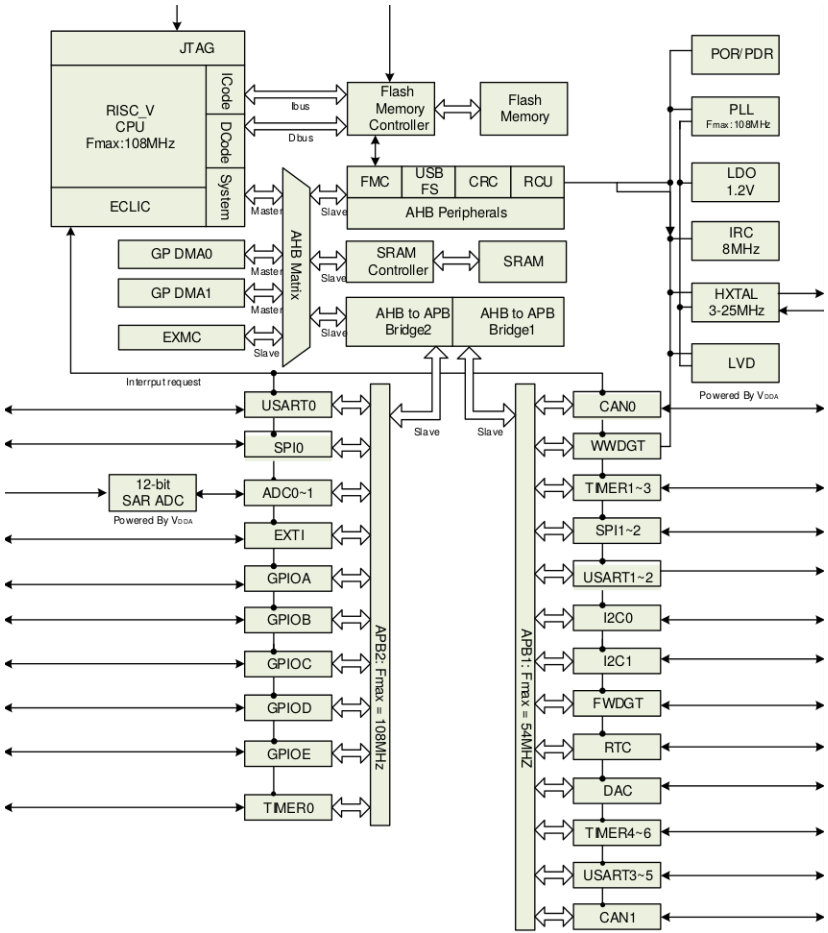
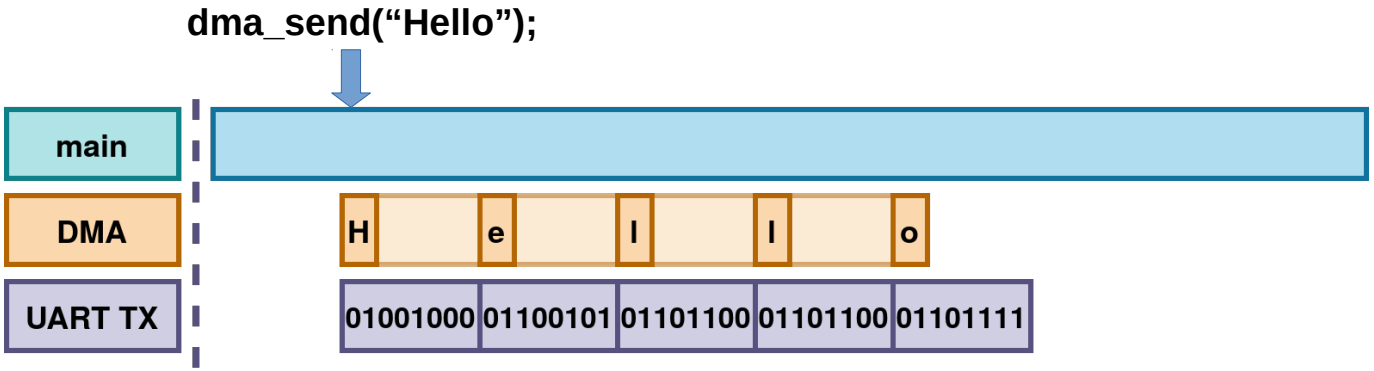
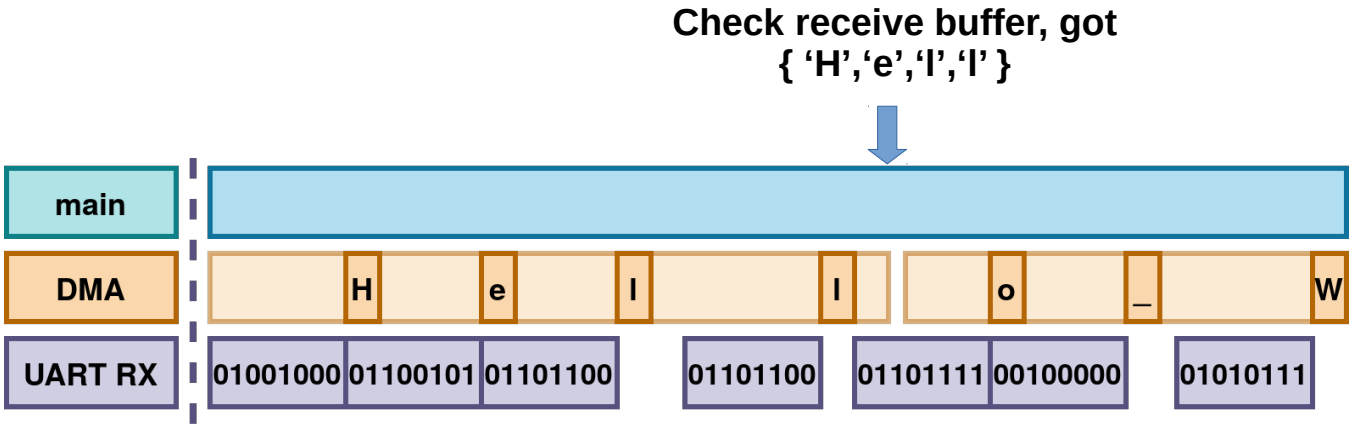
...provides a hardware method of transferring data between **peripherals** and/or memory...



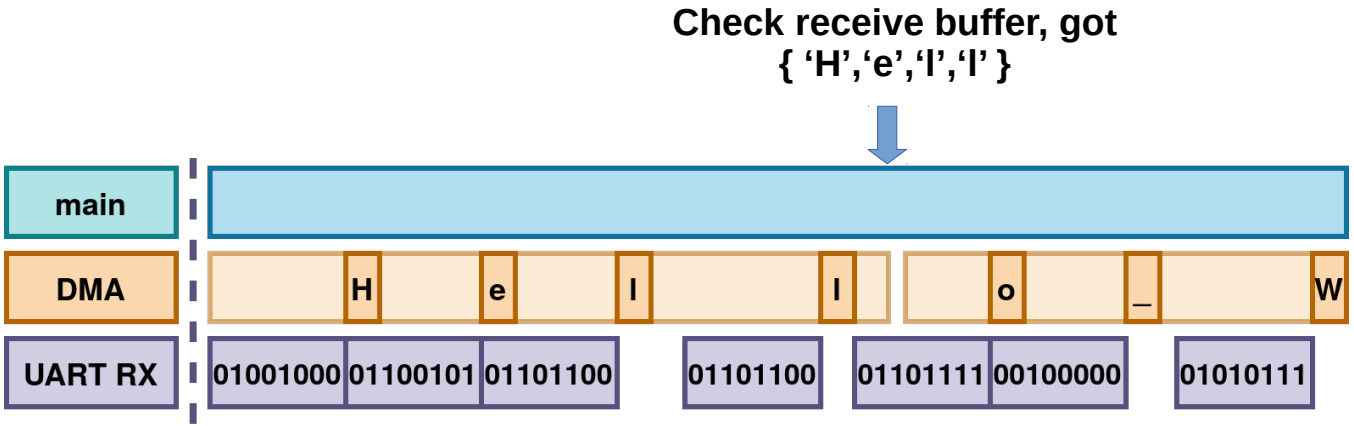
UART – Transmit buffer



UART – Receive buffer



UART – Receive buffer



Guarantees that we don’t miss any characters, unlike when polling!

UART – Example code

First initialize UART

```
/* USART configure */
usart_deinit(USART0);
usart_baudrate_set(USART0, 115200U);
usart_word_length_set(USART0, USART_WL_8BIT);
usart_stop_bit_set(USART0, USART_STB_1BIT);
usart_parity_config(USART0, USART_PM_NONE);
usart_hardware_flow_rts_config(USART0, USART_RTS_DISABLE);
usart_hardware_flow_cts_config(USART0, USART_CTS_DISABLE);
usart_receive_config(USART0, USART_RECEIVE_ENABLE);
usart_transmit_config(USART0, USART_TRANSMIT_ENABLE);
usart_enable(USART0);
```

Enable DMA

```
usart_dma_transmit_config(USART0, USART_DENT_ENABLE);
usart_dma_receive_config(USART0, USART_DENR_ENABLE);
```

UART – Example code RX

RX to RAM

```
dma_deinit(DMA0, DMA_CH4);
dma_init_struct.direction = DMA_PERIPHERAL_TO_MEMORY;
dma_init_struct.memory_addr = (uint32_t)rx_dma_buffer;
dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
dma_init_struct.number = LIO_BT_BUFFER_SIZE;
dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
dma_init(DMA0, DMA_CH4, &dma_init_struct);
    /* configure DMA mode */
dma_circulation_disable(DMA0, DMA_CH4);
    /* enable DMA channel4 */
dma_channel_enable(DMA0, DMA_CH4);
```

UART – Example code RX

Destination pointer,
Initialized with ‘\0’

```
dma_deinit(DMA0, DMA_CH4);
dma_init_struct.direction = DMA_PERIPHERAL_TO_MEMORY;
dma_init_struct.memory_addr = (uint32_t)rx_dma_buffer;
dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
dma_init_struct.number = LIO_BT_BUFFER_SIZE;
dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
dma_init(DMA0, DMA_CH4, &dma_init_struct);
    /* configure DMA mode */
dma_circulation_disable(DMA0, DMA_CH4);
    /* enable DMA channel4 */
dma_channel_enable(DMA0, DMA_CH4);
```

UART – Example code RX

```
dma_deinit(DMA0, DMA_CH4);
dma_init_struct.direction = DMA_PERIPHERAL_TO_MEMORY;
dma_init_struct.memory_addr = (uint32_t)rx_dma_buffer;
dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
dma_init_struct.number = LIO_BT_BUFFER_SIZE;
dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
dma_init(DMA0, DMA_CH4, &dma_init_struct);
    /* configure DMA mode */
dma_circulation_disable(DMA0, DMA_CH4);
    /* enable DMA channel4 */
dma_channel_enable(DMA0, DMA_CH4);
```

Peripheral address

UART – Example code RX

```
dma_deinit(DMA0, DMA_CH4);
dma_init_struct.direction = DMA_PERIPHERAL_TO_MEMORY;
dma_init_struct.memory_addr = (uint32_t)rx_dma_buffer;
dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
dma_init_struct.number = LIO_BT_BUFFER_SIZE;
dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
dma_init(DMA0, DMA_CH4, &dma_init_struct);
    /* configure DMA mode */
dma_circulation_disable(DMA0, DMA_CH4);
    /* enable DMA channel4 */
dma_channel_enable(DMA0, DMA_CH4);
```

UART – Example code RX

```
if(rx_dma_buffer[0] != '\0'){
    int i = 0;
    for(; i < LIO_BT_BUFFER_SIZE && rx_dma_buffer[i] != 0 && i < size - 1; i++){
        buffer[i] = rx_dma_buffer[i];
        rx_dma_buffer[i] = '\0';
    }
    buffer[i] = '\0';
}
```

UART – Example code RX

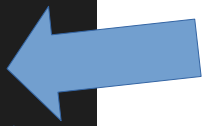
```
if(rx_dma_buffer[0] != '\0'){
    int i = 0;
    for(; i < LIO_BT_BUFFER_SIZE && rx_dma_buffer[i] != 0 && i < size - 1; i++){
        buffer[i] = rx_dma_buffer[i];
        rx_dma_buffer[i] = '\0';
    }
    buffer[i] = '\0';
}
```

Then initialize DMA again

UART – Example code

```
size_t lio_send_bt(char* message, uint32_t size)
{
    for(int i = 0; i < size && i < LIO_BT_BUFFER_SIZE; i++) {
        tx_dma_buffer[i] = message[i];
    }

    dma_parameter_struct dma_init_struct;
    dma_deinit(DMA0, DMA_CH3);
    dma_init_struct.direction = DMA_MEMORY_TO_PERIPHERAL;
    dma_init_struct.memory_addr = (uint32_t)tx_dma_buffer;
    dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
    dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
    dma_init_struct.number = 1;
    dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
    dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
    dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
    dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
    dma_init(DMA0, DMA_CH3, &dma_init_struct);
    /* configure DMA mode */
    dma_circulation_disable(DMA0, DMA_CH3);
    /* enable DMA channel3 */
    dma_channel_enable(DMA0, DMA_CH3);
}
```



Copy over to buffer

UART – Example code

```
size_t lio_send_bt(char* message, uint32_t size)
{
    for(int i = 0; i < size && i < LIO_BT_BUFFER_SIZE; i++) {
        tx_dma_buffer[i] = message[i];
    }

    dma_parameter_struct dma_init_struct;
    dma_deinit(DMA0, DMA_CH3);
    dma_init_struct.direction = DMA_MEMORY_TO_PERIPHERAL;
    dma_init_struct.memory_addr = (uint32_t)tx_dma_buffer;
    dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
    dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
    dma_init_struct.number = 1;
    dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
    dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
    dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
    dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
    dma_init(DMA0, DMA_CH3, &dma_init_struct);
    /* configure DMA mode */
    dma_circulation_disable(DMA0, DMA_CH3);
    /* enable DMA channel3 */
    dma_channel_enable(DMA0, DMA_CH3);
}
```

Initialize

UART – Example code

```
size_t lio_send_bt(char* message, uint32_t size)
{
    for(int i = 0; i < size && i < LIO_BT_BUFFER_SIZE; i++) {
        tx_dma_buffer[i] = message[i];
    }

    dma_parameter_struct dma_init_struct;
    dma_deinit(DMA0, DMA_CH3);
    dma_init_struct.direction = DMA_MEMORY_TO_PERIPHERAL;
    dma_init_struct.memory_addr = (uint32_t)tx_dma_buffer;
    dma_init_struct.memory_inc = DMA_MEMORY_INCREASE_ENABLE;
    dma_init_struct.memory_width = DMA_MEMORY_WIDTH_8BIT;
    dma_init_struct.number = 1;
    dma_init_struct.periph_addr = (uint32_t)&USART_DATA(USART0);
    dma_init_struct.periph_inc = DMA_PERIPH_INCREASE_DISABLE;
    dma_init_struct.periph_width = DMA_PERIPHERAL_WIDTH_8BIT;
    dma_init_struct.priority = DMA_PRIORITY_ULTRA_HIGH;
    dma_init(DMA0, DMA_CH3, &dma_init_struct);
    /* configure DMA mode */
    dma_circulation_disable(DMA0, DMA_CH3);
    /* enable DMA channel3 */
    dma_channel_enable(DMA0, DMA_CH3);
}
```

Enable

UART – Example code

```
size_t lio_send_bt(char* message, uint32_t size)
{
    for(int i = 0; i < size && i < LIO_BT_BUFFER_SIZE; i++) {
        tx_dma_buffer[i] = message[i];
    }

    dma_parameter_struct dma_init_struct;
    dma_deinit(DMA0, DMA_CH3);
    dma_init_struct.direction = DMA_MEMORY_TO_PERIPHERAL;
    dma_init_struct.memory_addr = (uint32_t)tx_dma_buffer;
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    dma_circulation_disable(DMA0, DMA_CH3);
    /* enable DMA channel3 */
    dma_channel_enable(DMA0, DMA_CH3);
}
```
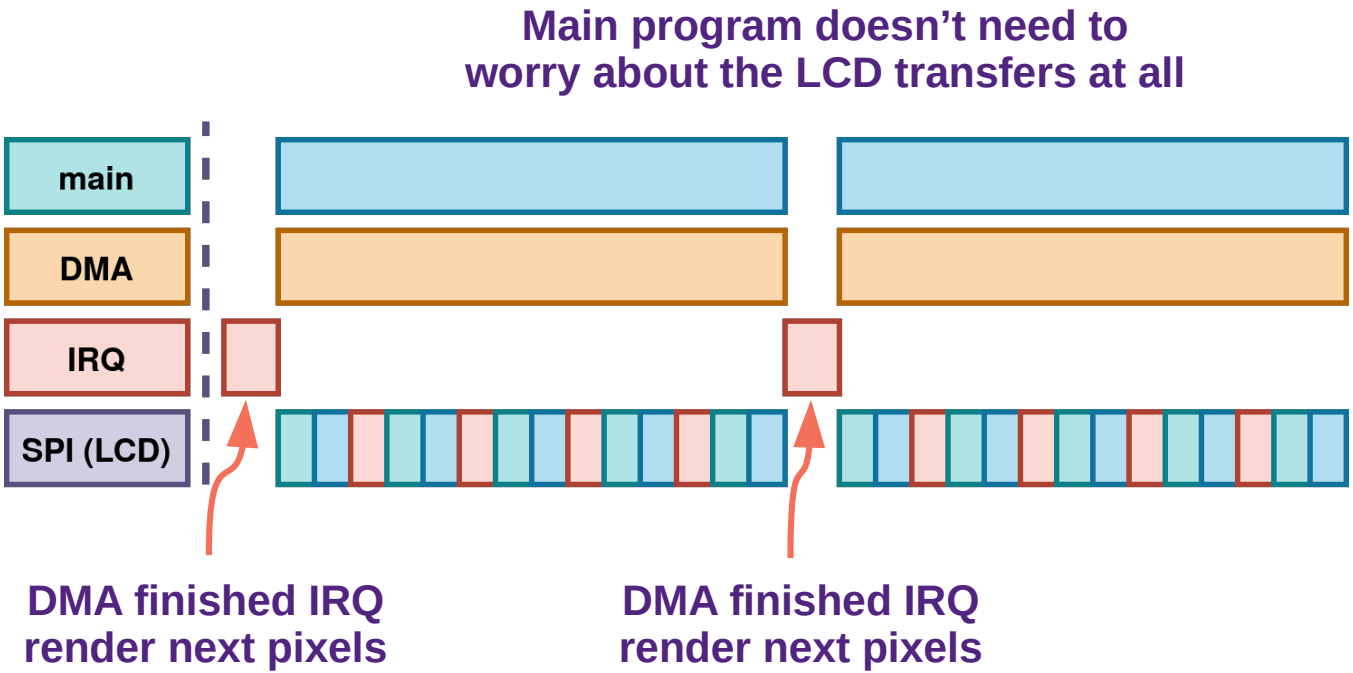

Enable

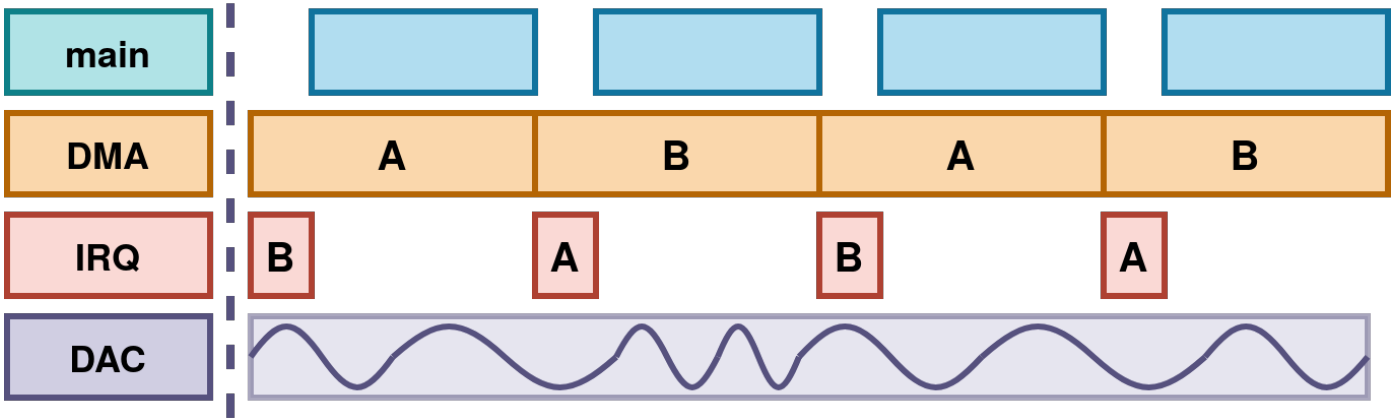
Message will go out automatically without any more code!

SPI – Callback interrupt



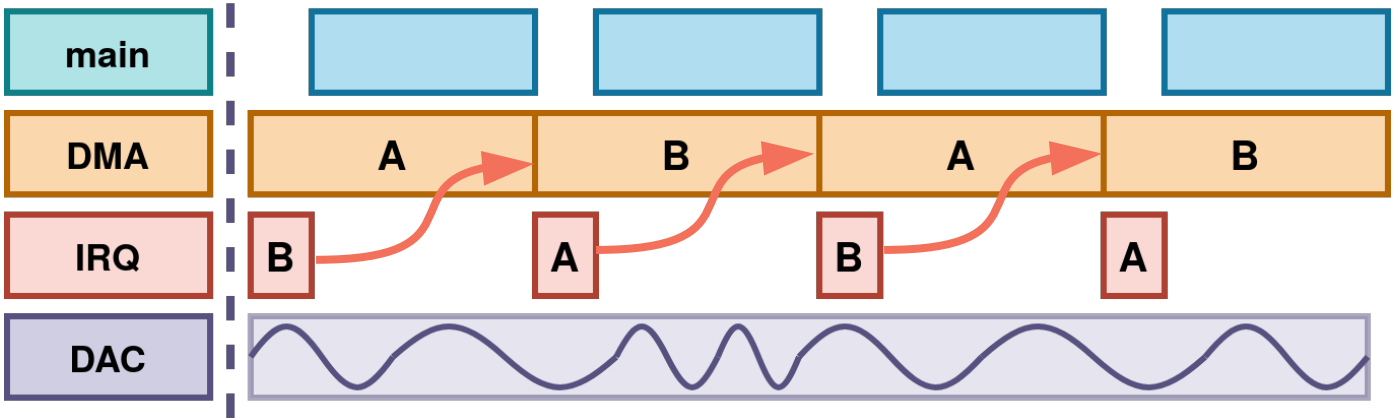
DAC – Double buffer

* AKA Ping Pong buffer,
or Page turn buffer



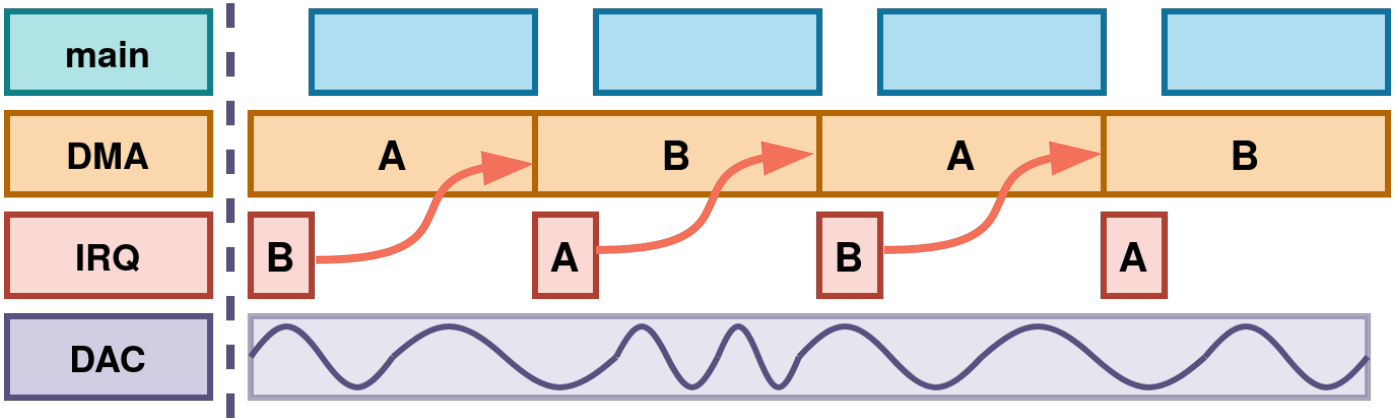
DAC – Double buffer

* AKA Ping Pong buffer,
or page flipping



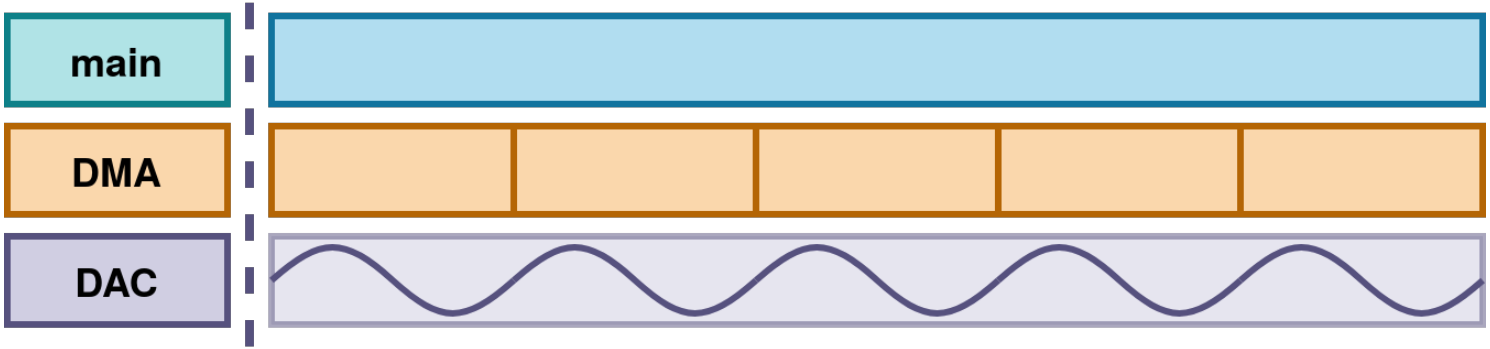
DAC – Double buffer

* AKA Ping Pong buffer,
or Page turn buffer



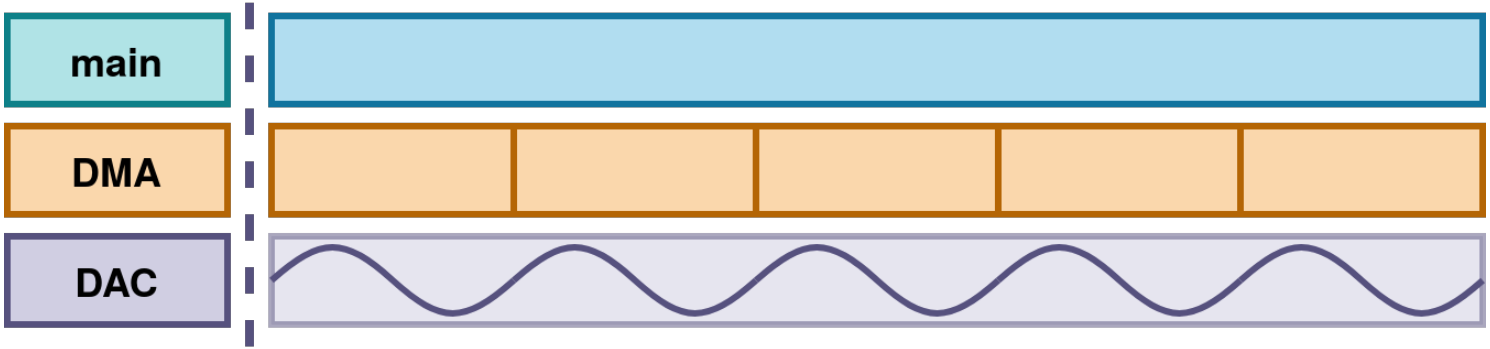
Introduces some latency and memory in exchange for consistency between samples.

DAC – Continuous



DAC – Continuous

“Set and forget”



Essentially zero processing overhead!

DMA - summary

For:

DMA - summary

For:

High bandwidth

DMA - summary

For:

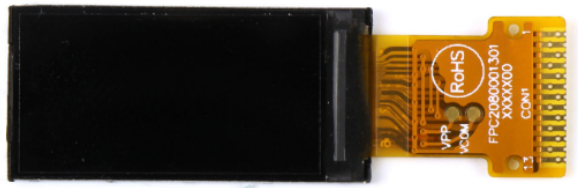
High bandwidth

Sporadic events

DMA - summary

For:

High bandwidth



Sporadic events

Timing sensitive